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Tracking innovation diffusion: AI analysis of large-scale patent data towards an agenda for further research

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Highlights

- To understand firm performance, it is important to study the effects of technology.
- AI-based algorithms are useful in producing data for studying technology trends.
- Specifically, DBSCAN algorithms are appropriate in that regard.

Abstract

Recent advances in AI algorithms and computational power have led to opportunities for new methods and tools. Particularly when it comes to detecting the current status of inter-industry technologies, the new tools can be of great assistance. This is important because the research focus has been on how firms generate value through managing their business models. However, further attention needs to be given to the external technological opportunities that also contribute to value creation in firms. We applied unsupervised machine learning techniques, particularly DBSCAN, in an attempt to generate a macro-level technological map. Our results show that AI and machine learning tools can indeed be used for these purposes, and DBSCAN is a potential algorithm. Further research is needed to improve the maps and to use the generated data to study related phenomena including entrepreneurship.

Introduction

A central question in research concerns how organizations adjust to innovation and technological uncertainty. A particular focus on tracking the technology trends in innovation has spawned significant interest due to its implications for future performance. All industries undergo changes in their technological base, which impacts how business models are defined and executed (Rothaermeland Hill,2005; Sirmonetal., 2008; Teece,2006). The logics or frameworks that generate profits by creating value for customers is often gradually but fundamentally changed over time. This has implications for how products and services are distributed by technology and for the long-term survival of organizations (Zottetal., 2011; Zottand Amit,2008). While both start-ups and incumbent firms respond to the emergent changes by altering the technological base on which they deliver their products and services (Rothaermeland Boeker,2008; Sirmonetal., 2008), the current focus has been confined to the business model itself (Chesbrough,2010; Chesbroughand Rosenbloom,2002). Far less attention has been paid to the underlying changes, to the technology trends that are emerging and altering the fundamentals of how products and services are created. Many firms change the business models and the technologies they employ with a view to capturing the potential value. What, however, are the early trends that both start-ups and incumbents respond to?

This is a challenging research question. Tracking technological developments can help to identify opportunities for venturing into new industries by providing knowledge about market trends. While some industries are mature and even declining, others are just emerging. In the early stages, the exact industry is hard to define. Knowledge about early changes in technological developments that reflect emerging and mature trends in industry is sparse and widely under- researched. This lack of scientific attention to tracking technological trends and the use cases of modern AI and machine learning has constricted our understanding of the past and current state of technology (Leeetal., 2009b; Yoon,2008) and the fundamental basis for entrepreneurial opportunities (Anokhinetal., 2011). And, it has constrained our knowledge of the fundamentals that underly business models (Chesbrough,2010).

The advent of extensive data availability, data analysis techniques, and technologies such as Artificial Intelligence (AI) has provided opportunities for research in this area. The opportunity lies in utilizing large-scale data on patent texts to track technological trends. Scholars have begun to utilize available data on patents in order to track these technological trends (Tsengetal., 2007). This approach, until recently, was limited because of restrictions in computing power and analysis techniques. These restrictions, however, are being alleviated, opening up opportunities to use AI for tracking technological trends and investigating the underlying logics for wider research on strategy, entrepreneurship, business models, and innovation. So far, different techniques have been used to obtain insights from patent bibliographic data (e.g., Leeetal., 2011). They include conventional techniques and unsupervised machine learning, which is an instance of AI-related techniques. For example, K-means clustering has been implemented to track technological trends (Suominenetal., 2017). However, analysis has yet to be forthcoming on how more sophisticated AI can be used to predict and track technology trends that start-ups and incumbents could employ to deliver value.

The objective of this article is to add to current understanding of how AI can be used to track technological trends. To do so, we use the titles of 16,384 patents for each year from 1976 to 2017. These were obtained from the USPTO[1]'s website. We seek to advance the proposition that DBSCAN[2] is an AI method that can be used specifically to track technological trends for the purpose of assisting the decision-making process regarding technological opportunities. In our research, we review the use of AI and machine learning in the innovation literature, and we illustrate how AI c be used to study an area that has received scant attention in previous research.

This paper's contribution is threefold. While previous research has provided theoretical grounds for studying the changing technological landscape, we propose a way to measure and track the changes. Since AI is finding its way in entrepreneurship research, we posit that DBSCAN is an appropriate algorithm for clustering patent titles so that technological trends can be tracked. Finally, we propose a mechanism through which new data can be generated. Such data could be used to study how these underlying technology trends are emerging and changing. This is particularly important since these studies suffer from a lack of data and, therefore, are rare in comparison to those that focus on business models.

In the remaining sections, we review the relevant literature concerning technological trends, decision making, and AI. Then, we explain our data and AI-based methodology. Finally, after illustrating the results, we draw our conclusions.

Section snippets

Tracking technological trends

It has been well established in the literature that technologies and industries have life cycles (Leeetal., 2016). New technologies emerge and, after early adoption, they go through a stage of growth until the point of maturity and saturation is reached (Ernst,1997). Their emergence partly stems from knowledge that is diffused from other technologies and other geographical places (Changetal., 2009). The theory of technology diffusion, in particular, implies such diffusions and life cycles. …

Data source and data approach

We utilize textual data on patent titles for the purpose of unsupervised machine learning. USPTO provides freely available data on more than 7 million patents over a 40-year period in the USA.

Explicitly, we collect Patent Grant Bibliographic (Front Page) Text Data (1976 - 2017) (cf. Lafondand Kim,2019). The data is available in various formats including the range of dat and xml files. We are interested in data in the patent titles in order to find clusters of technology. Accordingly, we took …

AI methods and algorithm

One way to cluster the information for the purpose of spotting new categories is to follow unsupervised machine learning techniques. Titles of more than 7 million patents provide a large amount of data that can be used in unsupervised learning. While processing the titles, we applied unsupervised text clustering techniques (Nigametal., 2000) namely DBSCAN (Esteretal., 1996). DBSCAN algorithm clusters those points in space that are packed closely together. It excludes outliers that are not …

Results

We fed the DBSCAN algorithm with quantified words (using tf-idf) from the patent titles in each year. The algorithm grouped together the words into a handful of clusters. The results, therefore, include 42 sets of clustered words related to the years included in the analysis. Each of the sets, representing one year, involves one cluster for excluded noise and the main clusters. Each cluster illustrates words (cleaned versions) that are common in patent titles in the respective cluster. We saved …

Discussion

The literature on management has widely studied business models (Zottetal., 2011). Furthermore, it has been argued that technological change provides opportunities to introduce new products, services, and business models (Rothaermeland Hill,2005; Sirmonetal., 2008). The aim of this study is to continue the discussion in the literature by proposing that AI, and specifically DBSCAN, can be utilized to map technological trends with the purpose of facilitating decision making concerning the …

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CRediT authorship contribution statement

Ashkan Fredström: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Joakim Wincent:** Conceptualization, Supervision, Validation, Writing - review & editing. **David Sjödin:** Conceptualization, Writing - review & editing. **Pejvak Oghazi:** Conceptualization, Writing - review & editing. **Vinit Parida:** Conceptualization, Writing - review & editing. …

Ashkan Fredström (Ph.D.) is affiliated researcher at Hanken School of Economics, Finland. ...

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AI-driven business model innovation: A systematic review and research agenda
2024, Journal of Business Research

> *Citation Excerpt :*
> …Establishing agile and flexible approaches, along with democratizing AI to involve the workforce in BMI activities, can overcome these cultural deficiencies (Korherr & Kanbach, 2021; Mao et al., 2021; Marinakis et al., 2021). AI-initiative rollouts should consider industry specifics and the size of the organization (Eling et al., 2022; Fredström et al., 2021; Trenerry et al., 2021; Volberda et al., 2021). Different types of organizations, such as startups and incumbents, have varying levels of adaptability and accessibility to data (Garbuio & Lin, 2019; Zaki, 2019)….

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> …Other researchers such as Lee and He (2021), Sjödin et al. (2018), Li et al. (2020) and Garbuio and Lin (2019) focus on problem recognition proposing AI as an innovative approach based on real data and evidence, while Maghrabie et al. (2019) discussed problem recognition by going back to theories on complex decisions and uncertainty. Fredström et al. (2021) looked to opportunity recognition, illustrating how AI tracks technological opportunities and trends. The second theme (12 papers) illustrates how AI unlocks innovation determinants in organizations, such as creativity, collective intelligence, and absorptive capacity.…

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